

Paul Stanley Ganganapalli

+49 162 7220780 | paulstanleyganganapalli@gmail.com | linkedin.com/in/paul-stanley-218002195 | github.com/PaulStanley0211 |
Stuttgart, Germany

PROFESSIONAL SUMMARY

Junior AI Engineer building production GenAI applications. 6 shipped projects spanning RAG architecture, LLM integration into enterprise systems, and DSGVO-compliant conversational AI. FastAPI + Claude + Docker stack across all production work; reproducible LLM evaluation with LLM-as-judge scoring. Three years translating market views into rule-based systems with audit trails - the same translation craft now applied to LLM use cases for enterprise clients. Open to consulting roles Germany-wide.

CORE COMPETENCIES

Generative AI

LLM Integration

Enterprise Architecture

RAG

Cloud Deployment

DSGVO

Stakeholder Translation

Python + SQL

WORK EXPERIENCE

Independent Quantitative Trader Oct 2022 - Sep 2025

India (Remote)

- Active trader in Indian equity, derivatives, and commodities markets - developed rule-based systems for entry/exit, position sizing, and post-trade review across 3 years of consistent practice.
- Built personal frameworks for risk management - daily loss limits, position sizing constraints, and trade journaling - patterns later codified into QuantFlow's 7-rule security guardrail and audit-trail architecture.
- Applied systems thinking from disciplined trading (clear rules, audit trails, defined risk limits) to AI engineering practice - particularly evaluation harness design and security guardrails.

Mechanical Design Engineer (Part-time) Mar 2021 - Feb 2022

Balaji Transporters and Handlers, India

- Designed mechanical components and assemblies for material-handling equipment using SolidWorks and AutoCAD; produced 3D models, technical drawings, and assembly documentation for production.
- Conducted design reviews and revisions across the product lifecycle, optimizing for manufacturability and cost; collaborated with manufacturing and quality teams to validate feasibility before production.
- Applied engineering judgment to troubleshoot design issues during prototyping and pre-production, contributing to reliable production-ready outputs.

PROJECTS

Production RAG System RAG · Hybrid Retrieval · LLM Evaluation

Production-grade RAG with hybrid retrieval, agentic CRAG, three-layer security, and a reproducible LLM-as-judge evaluation harness. Built end-to-end RAG pipeline: hybrid retrieval (BGE-small dense + BM25 sparse with Reciprocal Rank Fusion), cross-encoder reranking, CRAG self-correction. Implemented three security layers: prompt-injection input guard, content filter on retrieved chunks, post-stream PII redaction with Luhn-validated credit-card detection plus email/phone/SSN regex. Achieved **Hit Rate@5 = 0.96, MRR = 0.96, faithfulness = 0.944, 100% routing accuracy** on a reproducible eval harness (30 retrieval cases, 15 generation cases, LLM-as-judge scoring). Architected SSE streaming API with token/citation/status events; tiered model strategy (Sonnet for generation, Haiku for routing/grading) keeping per-query cost in low single cents.

Python · FastAPI · Qdrant · Redis · Anthropic Claude · React · TypeScript · Docker · github.com/PaulStanley0211/Production_RAG_System

MIA - German Retail AI Agent Conversational AI · DSGVO · Bilingual

Production-grade conversational AI for German e-commerce with bilingual support and DSGVO-compliant memory. Architected full-stack AI system managing 25+ orders and 25+ products across 6 categories using PostgreSQL persistence and LangChain orchestration with Claude as the reasoning engine. Automated 5 core customer-service functions (order tracking, returns, product recommendations, FAQ, complaints), reducing support workload by **60-70%**. Implemented persisted vector memory in ChromaDB enabling personalized customer context across sessions, with automatic DSGVO-compliant cleanup of expired user data. Engineered bilingual support (German + English) with automatic language detection, deployed via Docker with FastAPI REST API (3 endpoints) and Streamlit web interface for the German e-commerce market.

Python · Anthropic Claude · LangChain · PostgreSQL · FastAPI · ChromaDB · Streamlit · Docker · github.com/PaulStanley0211/MIA---German-Retail-Agent

QuantFlow - Multi-Agent Trading Workflow for German DAX 40 Multi-Agent · HITL · Audit Trail

8-agent orchestrated system for DAX 40 pre-market analysis, backtesting, and human-in-the-loop trade approval. Architected 8 specialized agents (morning briefing, setup scanner, risk monitor, trade executor, security guardrails, strategy builder) with scheduled triggers, email-based human approval workflow via IMAP, and file-based audit trail for full traceability. Built full-stack Strategy Builder: FastAPI backend with SSE streaming, React + Recharts dashboard, plain-English-to-JSON strategy parser via Claude, multi-instrument backtest engine across 54 instruments (DAX 40, indexes, crypto, commodities). Implemented 7-rule pre-trade security guardrail (stop-loss verification, position sizing, R/R minimum, daily loss limit, trade count cap, blocklist, duplicate detection) with hard-block on rule violations. Designed paper-trading-only with no broker integration - every trade requires human YES/NO email approval before logging, ensuring full auditability and zero automated execution risk.

Python · Anthropic Claude · FastAPI · React · Recharts · yfinance · IMAP/SMTP · github.com/PaulStanley0211/Quantflow

EDUCATION

Master of Engineering Science in Mechanical Engineering - Riga Technical University, Riga, Latvia	Feb 2019 - Feb 2021
Bachelor of Technology in Mechanical Engineering - Karunya Institute of Technology and Sciences, Coimbatore, India	Jun 2014 - Jun 2018

CERTIFICATIONS

Microsoft Azure Fundamentals	AZ-900
Microsoft Azure AI Fundamentals	AI-900

SKILLS

Languages & Frameworks: Python, SQL, TypeScript, FastAPI, React, LangChain, LangGraph, Streamlit

AI Engineering: RAG, Hybrid Retrieval, Agentic Workflows, Vector Search, Cross-encoder Reranking, LLM Evaluation, Prompt Engineering, LLM-as-Judge, Multi-agent Orchestration

LLMs & APIs: Anthropic Claude (Sonnet, Haiku, Opus), Claude Vision, OpenAI API

Data & Storage: PostgreSQL, SQLite, Qdrant, ChromaDB, FAISS, Redis

DevOps & Infrastructure: Docker, Docker Compose, Git/GitHub, Server-Sent Events (SSE), Azure

Languages (spoken): English C1 (Advanced), German B1 (Intermediate, working towards B2)